CHAPTER 10

Causal Inference for Time Series

10.1 Introduction and Core Concepts

10.1.1 Time Series vs. Cross-Sectional Causality

Causal inference in time series data differs fundamentally from cross-sectional approaches. The distinguishing feature is **temporal structure**: in time series, we observe the same units repeatedly over time, creating opportunities and challenges absent in cross-sectional designs.

Univariate time series causal inference compares a unit to **itself** over time. For example, economists might compare a city's crime rate **before** and **after** a new policing policy. The unit serves as its own control, but this creates a critical complication: observations are **not** independent. Each observation depends on the previous one through autocorrelation, violating the independence assumption of standard statistical tests.

By contrast, **cross-sectional causal inference** compares different units at a single point in time—for instance, comparing treatment outcomes between patients in one cohort. Cross-sectional designs assume units are (conditionally) exchangeable, which is straightforward but requires many comparison units.

Time series designs exploit what we might call the **arrow of time**: causes must precede effects. This principle eliminates backward arrows from causal graphs a priori, pruning half the possible causal directions. However, time series designs face distinct challenges. Observations are correlated across time, treatments and confounders may drift or trend, and experiments are often impossible. A canonical example illustrates the difficulty: did the Federal Reserve rate cut in 2020 cause the stock market rebound? We cannot rerun 2020 with a different policy, so we must rely on temporal structure and domain knowledge to reason counterfactually.

10.1.2 Temporal Causal Structures: Basic Setup

In time series causal models, we typically observe sequences of variables over time: a treatment process T_t (continuous or binary), an outcome Y_t, and covariates X_t. The fundamental principle is that **effects cannot precede causes**. This means treatment at time t can affect outcomes at times t, $t + 1$, $t + 2$, and beyond, but never at times before t. Mathematically, T_t can cause Y_t, $Y_{\{t+1\}}$, etc., but never $Y_{\{t-1\}}$.

This constraint is immensely powerful because it eliminates a large class of incorrect causal directions without any data analysis. We can represent this temporal structure in a **temporal causal graph**. Figure 1 illustrates a simple first-order graph showing how variables at consecutive time points relate. The restriction that T_t affects Y_t but not prior outcomes is encoded by the arrow structure.

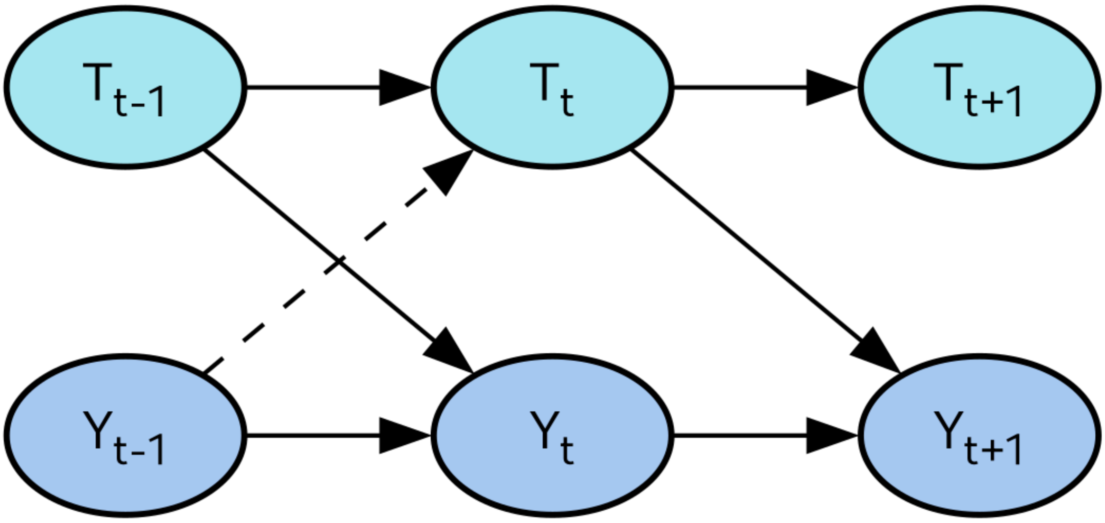

Figure 1: First-order temporal causal graph showing treatment and outcome

10.1.3 Temporal Causal Structures: Key Quantities

The central causal quantity in time series is the effect of a **sustained intervention**. Suppose we force the treatment sequence from some time t_start onward to a fixed level. The **sustained intervention causal effect** is defined as:

The causal effect of a sustained intervention is the difference in expected outcomes between forcing treatment on versus off from time t_start onward.

This measures the average outcome at time t when we force the treatment to one level versus zero from t_0 onward. The intuition is that we **force** the treatment sequence, contrasting it with what would have happened without intervention.

Dynamic treatment occurs when treatment at time t depends on the unit's history: covariates $X_{\{1:t\}}$, past outcomes $Y_{\{1:t-1\}}$, and past treatments $T_{\{1:t-1\}}$. This is common in medicine, where dosing depends on patient response so far.

Two special cases are particularly important. **Contemporaneous effects** occur when $T_t to Y_t$ instantly. **Lagged effects** occur when for some lag $k > 0$. In monetary policy, for example, a rate cut this quarter affects GDP over several subsequent quarters (a lagged effect). Simultaneously, the Fed adjusts rates based on past GDP, creating feedback loops that complicate causal inference.

10.2 Challenges Specific to Time Series

10.2.1 Autocorrelation and Standard Errors

Autocorrelation means that Y_t is correlated with its own past values: $Y_{\{t-1\}}$, $Y_{\{t-2\}}$, etc. This has profound consequences for statistical inference. When we compute standard errors assuming independent observations, our estimates are **too small**. Autocorrelated data carries less independent information than the number of observations suggests. Significance tests reject the null hypothesis far more often than their nominal rate, leading to spurious conclusions.

Consider a simple example: regressing daily stock returns on a dummy variable for "news day." Stock returns exhibit strong autocorrelation. If we ignore this and apply standard OLS, we compute incorrect standard errors, likely concluding that news has a statistically significant effect when it does not.

The solution is to use **Newey-West or HAC** (heteroskedasticity- and autocorrelation-consistent) standard errors. These estimators account for correlation at multiple lags and asymptotically produce correct inference. When autocorrelation is present, always use HAC standard errors, cluster standard errors, or bootstrap methods rather than naive OLS inference.

10.2.2 Non-Stationarity and Trends

A **stationary process** has constant mean, variance, and autocovariance over time. Many real-world economic and social time series are **not** stationary; GDP, prices, and populations grow, creating trends that persist.

Non-stationarity leads to **spurious regression**, a classic pitfall. Two entirely unrelated series that happen to trend upward will appear highly correlated even though one does not cause the other. The example of pirates and global temperature is famous: the number of pirates has declined over two centuries while global temperature has increased, creating a strong negative correlation—yet clearly one does not cause the other.

Several solutions exist. We can **difference the series**, working with the first difference instead of levels. We can **detrend** by removing a fitted trend before analysis. If two series share a common stochastic trend, we can use **cointegration methods** that model their long-run relationship while controlling for the shared trend.

The key is to test for stationarity using unit-root tests (ADF, KPSS) and transform non-stationary series before causal inference.

10.2.3 Feedback Loops and Simultaneity

In time series, cause and effect frequently influence each other, creating **feedback loops**. This causes **simultaneity bias** in regression: when Y_t and T_t are jointly determined by unobserved shocks, OLS estimates are inconsistent, and coefficients cannot be interpreted as causal effects.

A clear example is monetary policy and inflation. The central bank cuts rates to stimulate the economy, which increases inflation. Policymakers then raise rates to combat inflation, which dampens growth. The policy (T_t) affects inflation (Y_t), but inflation simultaneously influences future policy. This reciprocal relationship invalidates naive causal inference, as illustrated in Figure 2.

Another example is advertising and sales. Companies increase advertising budgets when sales rise (or to recover from low sales). Sales rise partly due to advertising and partly due to external demand shocks. Advertising in turn responds to sales, creating bidirectional causality that OLS cannot untangle.

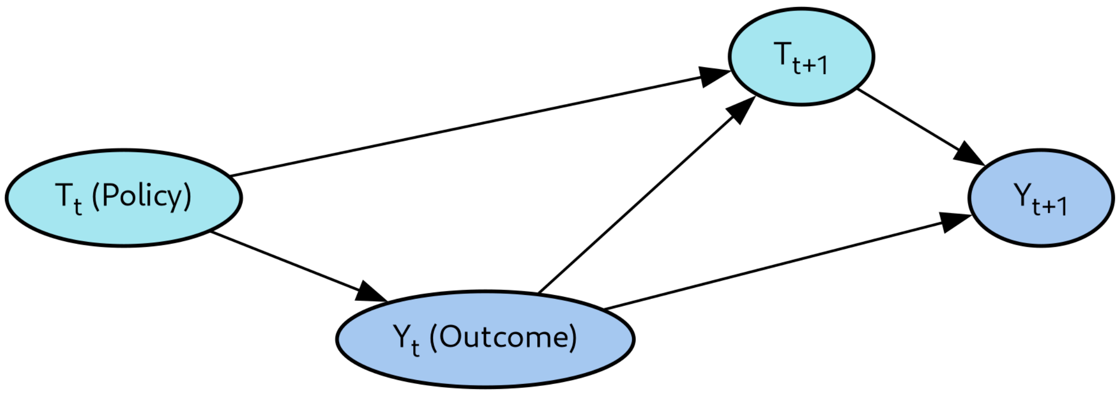

Figure 2: Feedback loop structure where policy affects outcome contemporaneously,

10.2.4 Structural Vector Autoregressions

Structural Vector Autoregressions (VARs) offer one approach to feedback loops. Rather than treating policy and outcome asymmetrically (as in observational causal inference), VAR models treat all variables symmetrically, estimating how they influence each other over time.

A **VAR(p) model** with k variables is formulated as:

Here vY_t is a vector of k time series, the A_i are coefficient matrices encoding dependence on lagged values, and ε_t are structural shocks (uncorrelated across equations). The VAR estimates how shocks propagate through the system, revealing the causal effects of unexpected changes.

The key challenge is **identification**: how do we determine causality direction? The solution is to impose **structural assumptions**. For instance, in monetary policy, we might assume that the Fed reacts to inflation **within** the same quarter, but inflation responds to policy **in the next quarter**. This timing restriction identifies the causal direction. Once identified, we can simulate interventions (permanent shocks) and trace their effects through the system.

10.2.5 Instrumental Variables for Time Series

Instrumental Variables (IV) break feedback loops by providing exogenous variation in treatment. The key idea is finding a variable Z that affects treatment T but not outcome Y directly (the **exclusion restriction**). We use Z to predict T, obtaining , then measure how correlates with Y.

This approach works because Z provides variation in treatment independent of confounds, allowing us to isolate the causal effect. In education, scholars study the effect of schooling on earnings using distance to the nearest college as an instrument: distance affects educational access but not earnings directly (barring unusual circumstances).

The limitation is that IV estimates effects only on **compliers**—units whose treatment changes with the instrument. Compliers may differ from the full population, limiting generalizability. Additionally, finding a credible instrument is difficult in practice.

10.2.6 Natural Experiments and Exogenous Shocks

Natural experiments exploit unexpected external events that randomly assign treatment to some units but not others, mimicking a randomized trial in observational data. The advantage is that randomization eliminates confounding without requiring an instrument.

A famous example is the Vietnam War draft lottery, which randomly assigned military service. Researchers compared drafted and non-drafted veterans on lifetime earnings, estimating the causal effect of military service. Because the lottery was random, we can attribute earnings differences to military service, not selection effects.

Natural experiments are credible when the shock is truly exogenous (outside the system's control) and affects treatment assignment but not outcomes directly. However, they are rare, cannot be replicated, and typically estimate effects in specific contexts that may not generalize.

10.2.7 Time-Varying Confounders

Time-varying confounders make causal identification harder. When a confounder U_t affects both treatment T_t and outcome Y_t and changes over time, it contaminates causal estimates.

Consider evaluating a diet on weight loss. Season (winter vs. summer), stress, and motivation all change over time and drive both adherence and weight change. These are hard to measure but crucial confounders. Naive before-after comparisons will be biased by these time-varying factors.

The solution depends on whether confounders are time-invariant or time-varying. If a confounder is **time-invariant** (e.g., genetics), simple before-after comparisons can difference it out. This is the core principle behind fixed-effects models and Difference-in-Differences. If the confounder is **time-varying** (e.g., mood, weather), differencing does not eliminate bias. More sophisticated methods are needed: modeling the confounder explicitly, proxying for it with measured variables, or finding exogenous variation unrelated to the confounder.

10.2.8 Other Practical Challenges

Several additional complications arise in time series causal inference:

Anticipation effects occur when units change behavior **before** treatment if they expect it. When a tax cut is announced but not yet implemented, consumers may delay purchases to time them with the cut, biasing estimates.

Spillovers and SUTVA violations happen when treatment on one unit affects another. If advertising in one region spills to neighboring regions, we cannot isolate the effect to the treated region.

Structural breaks occur when the data-generating process changes mid-sample. The 2008 financial crisis and COVID-19 pandemic are extreme examples, but smaller breaks (policy regime changes, technological shifts) are common. Models fit before the break may not hold afterward.

Missing data and irregular sampling arise because standard time series tools assume equispaced observations. Sensor dropouts, holidays, weekends, and other irregularities complicate analysis. Imputation or specialized methods (e.g., state-space models) are needed.

10.2.9 When Temporal Structure Helps

Temporal structure is a resource for causal inference when exploited correctly. Time orders causes before effects, eliminating many causal directions a priori. This reduces the number of hypotheses and focuses analysis on plausible mechanisms.

The unit serves as its own control, so **time-invariant confounders** are automatically controlled. A person's genes, geography, and institutional features remain constant over time, so within-person comparisons difference them out. This is much more powerful than requiring a separate control unit.

Natural experiments are easier to spot in time series. Policy changes, shocks, and discontinuities provide quasi-experimental variation. Interrupted Time Series, Difference-in-Differences, and synthetic control all exploit this variation. When studying the effect of a sugary-drink ban, for example, we compare the same city before and after. Genes, climate, and culture are held fixed, making the comparison much cleaner than comparing different cities.

10.2.10 When Temporal Structure Misleads

Conversely, temporal structure can mislead. Trends fake causation: two series trending upward look causally linked even if unrelated (spurious regression). Seasonality hides effects when a seasonal bump masks or amplifies a policy effect.

Lagged effects confuse attribution. Today's outcome may reflect yesterday's treatment, so wrong lag specification produces wrong effect estimates. Feedback inflates correlations: if Y_t drives and T_t drives Y_t, regressing Y_t on T_t picks up both the treatment effect and the reverse effect, confounding inference.

Regime changes invalidate models. A model fit before COVID may not generalize after. The key takeaway is that temporal data is informative, but blindly applying cross-sectional tools to it creates more problems than it solves.

10.3 Granger Causality

10.3.1 Intuition and Definition

Granger Causality, named after Nobel laureate Clive Granger, answers a specific question: "Does knowing the past of X help me predict Y beyond what Y's own past already tells me?" The intuition is that causes must come before effects, and effects should carry information about their causes.

It is important to understand what Granger causality is NOT. It is not causation in Pearl's do-calculus sense (i.e., causal effect in the interventional sense). Rather, it is a predictive notion based on temporal precedence. Granger causality answers "does X help predict Y?" not "does changing X change Y?" An important consequence is that both $XtoY$ and $YtoX$ can hold simultaneously in Granger causality, which is impossible in deterministic causality but common in economic feedback systems.

A famous example illustrates the limitation: ice cream sales do not Granger-cause drowning. Instead, **temperature** Granger-causes both ice cream sales and drowning. They are correlated because of a common cause, not because one causes the other. Granger causality captures this dependence but misses the underlying mechanism.

10.3.2 Formal Definition

Consider two stationary time series X_t and Y_t. We compare two predictors of Y_t:

1. **Restricted model**: uses only past Y

2. **Unrestricted model**: adds past X

Definition: X **Granger-causes** Y if the unrestricted model has **strictly smaller** forecast error variance than the restricted model. Equivalently, an F-test on $H_0 : b\eta_1 = b\eta_2 = c... = b\eta_p = 0$ rejects, meaning at least one past value of X predicts current Y significantly.

The test is straightforward: fit both models by OLS, compute sum-of-squared residuals, and test whether including X reduces the residual sum of squares significantly. If yes, we conclude X Granger-causes Y.

10.3.3 Key Assumptions

Granger causality relies on several assumptions, violations of which lead to spurious conclusions. **Stationarity** is essential: both series must have time-invariant statistical properties. Non-stationary series can produce spurious Granger causality. The remedy is to pre-test with unit-root tests (Augmented Dickey-Fuller, KPSS) and difference series if needed.

Correct lag order p is critical. Too few lags miss the true effect, while too many lead to overfitting and loss of power. Lag order is typically chosen via information criteria (AIC, BIC) or cross-validation. Under- or over-specification biases tests.

Linearity is assumed in the basic formulation. Non-linear Granger causality extensions exist (e.g., transfer entropy and machine-learning based methods) but are beyond standard practice.

No omitted common causes—this is the "big one." If an unobserved variable Z drives both X and Y with different lags, X will appear to Granger-cause Y even though it does not. This cannot be tested; we must rely on domain knowledge.

Correct sampling frequency is needed. If the true effect operates faster than the sampling interval, Granger causality may fail to detect it or even reverse the direction.

10.3.4 Limitations

Granger causality is not causation. It tests **predictive** relationships, not **interventional** ones. Granger-causal relationships can be entirely spurious, driven by a common cause with no true causal link.

The barometer example is instructive: a barometer reading X_t Granger-causes storms because falling atmospheric pressure precedes storms. But manually raising the barometer does **not** cause a storm. The atmosphere is the true cause; the barometer merely signals it.

Aggregation problems arise when sampling frequency is wrong. Monthly data may miss daily Granger effects. If the process evolves faster than the sampling rate, we observe spurious bidirectional causality (each series Granger-causing the other) due to time aggregation.

Contemporaneous effects cannot be handled in the basic form. If X_t affects Y_t instantaneously (same period), lag-based tests miss it. Extensions exist (VAR formulations), but standard Granger causality tests skip contemporaneous relationships.

Long time series are required for reliable inference. Typically we need $Tggp$—for 4 lags, at least 100 observations. Short samples produce unreliable inference and low power.

10.3.5 Practical Example: Advertising and Sales

Consider the question: "Does advertising spend Granger-cause sales?" Suppose we have weekly data on ad spend X_t and sales Y_t. We fit both the restricted model (sales on past sales only) and unrestricted model (sales on past sales and past ads). If the unrestricted model has significantly lower forecast error, we conclude ad spend Granger-causes sales.

However, a critical caveat applies: this does **not** mean "if we spend more on ads, sales will rise." Both variables could be driven by seasonal demand patterns. A consumer goods company spends more on ads before holidays (which also see naturally higher sales), creating an apparent Granger relationship. To get an interventional answer—what happens if we **change** ad spending—requires an experiment or a causal design (e.g., randomized experiments, natural experiments).

Another widely-studied question is whether Twitter sentiment scores Granger-cause stock returns. Research finds mixed results: sentiment does Granger-cause short-horizon returns for some stocks, but effect sizes are small and may reflect transaction-level feedback loops rather than true predictive value.

10.4 Interrupted Time Series (ITS)

10.4.1 Design and Motivation

Interrupted Time Series (ITS) designs are used when a policy or intervention is implemented at a single known date t_start across all (or most) units. Examples include a new speed limit enforced nationwide from January 1st, or a website redesign launched on a specific day.

The **fundamental challenge** is the absence of a control group: the whole population is treated. A naive before-after comparison might show change, but we cannot distinguish the policy effect from concurrent trends or other events.

The **solution** is to use the **pre-intervention trajectory** as a counterfactual forecast. We fit a model to pre-intervention data, extrapolate it forward under the assumption that the trend would have continued absent the intervention, then compare the forecast (counterfactual) to what actually happened. The gap is the estimated causal effect.

ITS has impressive real-world examples. Seatbelt laws reduced traffic fatalities by approximately 15–20%. Netflix's password-sharing ban had mixed results: growth in some regions but churn in others, illustrating that counterfactuals are often uncertain. Apple's iOS privacy changes (requiring apps to request tracking permission) saw opt-in rates plummet from 60–70% to 15–25%, devastating ad targeting. Meta reported billions in revenue losses, demonstrating the power of identifying when policy changes affect outcomes.

As illustrated in Figure 3, the core ITS strategy is to extrapolate the pre-period trend as a counterfactual, then compare it to the actual post-period outcome. The gap represents the policy effect.

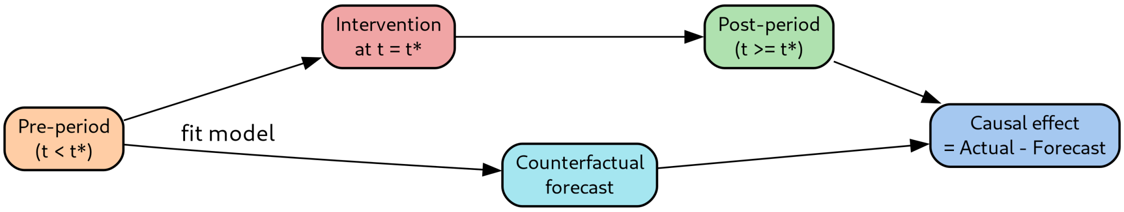

Figure 3: Interrupted Time Series design: using pre-period trends as counterfactual

10.4.2 Segmented Regression Formulation

The standard ITS model is **segmented regression**:

Y_t = beta_0 + beta_1 t + beta_2 D_t + beta_3 (t - t_0) D_t + u_t

where D_t is an indicator for the post-intervention period ($D_t = 1$ if $t \geq t^*$, and 0 otherwise).

The parameters have clear interpretations:
- $b\eta_0$: pre-intervention intercept (baseline level)
- $b\eta_1$: pre-intervention slope (trend before the intervention)
- $b\eta_2$: **level change** at the intervention (immediate jump)
- $b\eta_3$: **slope change** after the intervention (change in trend)

This formulation lets us test two key hypotheses:
1. Is there a jump at t_start ? Test $H_0 : b\eta_2 = 0$.
2. Does the trajectory change? Test $H_0 : b\eta_3 = 0$.

The estimated effect is $b\eta_2$ (immediate level change) plus the cumulative effect of $b\eta_3$ over the post-period (slope change). A positive $b\eta_2$ means the outcome jumped up at intervention; a positive $b\eta_3$ means the trend steepened.

10.4.3 Implementation Details

Several practical details matter for valid ITS:

Autocorrelation is almost always present in time series. Standard OLS standard errors are too small, leading to overconfidence. Fit models using autocorrelation-robust methods (Newey-West), ARIMA (AutoRegressive Integrated Moving Average), or other time-series methods that account for serial correlation.

Robustness checks strengthen ITS conclusions. Vary the pre-intervention window (use 1, 2, 5 years of data) to see if results are stable. Run placebo tests by pretending the intervention occurred at a fake date; if you find spurious effects, your design is picking up something other than the true intervention.

Always check for confounding events near t_start : seasonality, holidays, economic shocks. Visual diagnostics are crucial: plot the series with the fitted model and the intervention date marked. Do the ITS results survive inspection?

10.4.4 Applications

ITS is widely used across domains:

- **Public health**: estimating the effect of soda taxes on sugar-sweetened beverage purchases.
- **Policy evaluation**: measuring the effect of congestion pricing in London (2003) on traffic volumes, or minimum wage increases on employment.
- **Tech and product**: quantifying the effect of a website redesign on conversion rate or a new recommendation algorithm on engagement.
- **Safety interventions**: did installing speed cameras reduce traffic accidents?

Each application requires careful attention to the timing of the intervention, confounding events, and data quality.

10.4.5 Strengths and Weaknesses

Pros of ITS:
- Works when **no control group exists**: policies are often applied universally (same rules for all units).
- Uses the unit as its own control, so **time-invariant confounders cancel**: genes, geography, and institutional features are held fixed.

Cons of ITS:
- Relies on **extrapolation** of the pre-period trend, introducing model dependence.
- Vulnerable to **co-occurring events**: “was it the policy or the recession?”
- Autocorrelation inflates errors if ignored.

- Needs a long, stable pre-period, which is rare in practice.

Rule of thumb: ITS is most credible when:
1. The intervention is sharply timed (clear before/after boundary).
2. No other major events occur near t_start .
3. The pre-period is long and stable.
4. The outcome has low noise relative to effect size.

10.5 Difference-in-Differences (DiD)

10.5.1 Motivation and Setup

Motivating Example
- Suppose a policy is rolled out in **some** units but not others
  - E.g., New Jersey raises its minimum wage in 1992; Pennsylvania does not
  - Two groups: treated (NJ) and control (PA); two periods: before and after

Definition: Difference-in-Differences (DiD) estimates a causal effect by comparing the change in outcomes over time between a group that received treatment and a group that didn't
- Assume both groups would have followed the same trend in the absence of treatment

Procedure

- Subtracts out time-invariant differences between groups
- Subtracts out common time trends across groups
- What's left is attributed to the policy
- **Example**: Card & Krueger (1994) studied the NJ minimum wage
  - Standard theory predicted employment would fall
  - DiD estimate: essentially zero (a surprising result)

10.5.2 The Core Idea Graphically

The DiD estimator with two groups and two periods is shown in Figure 4. The estimator is: (mean treated post - mean treated pre) - (mean control post - mean control pre).

Here the treated group mean is compared against the control mean, with subscripts denoting time periods (pre vs. post). Visually, we compute the change in the treated group (pre to post) and subtract the change in the control group. The residual is the estimated causal effect, with the visual arrangement highlighting how both between-group differences and common time trends are removed.

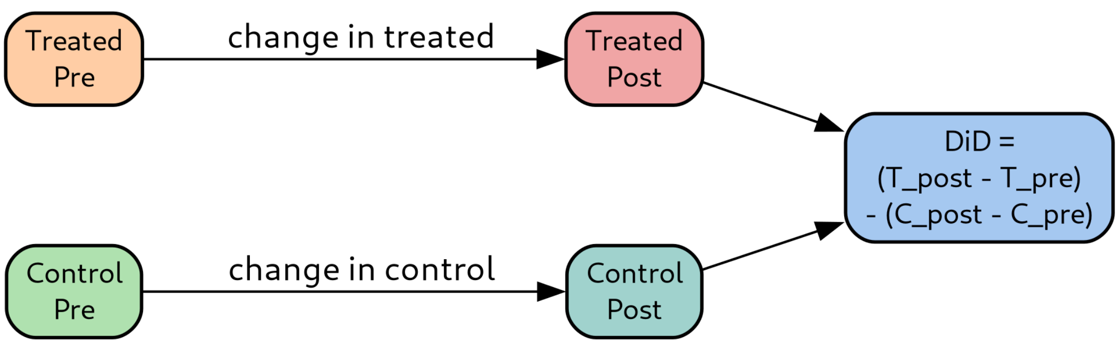

Figure 4: Difference-in-Differences design: the causal effect is the difference

10.5.3 Regression Formulation

The equivalent regression formulation with unit i and time t is:

DiD regression: $Y = \alpha + \beta T + \gamma P + \tau (T \times P) + u$

The parameters are:
- α: baseline outcome in control units in pre-period
- β: time-invariant difference between groups
- γ: common time trend affecting both groups
- τ: **the DiD causal effect** (coefficient on the interaction term)

A more flexible **fixed-effects** generalization is:

$$Y_{\{i,t\}} = \alpha_i + \lambda_t + \tau D_{\{i,t\}} + \varepsilon_{\{i,t\}}$$

Here α_i are unit fixed effects (absorb all time-invariant unit traits: location, management, size), λ_t are time fixed effects (absorb all common shocks: recessions, industry trends), and $D_{\{i,t\}}$ indicates whether unit i is treated at time t. The fixed-effects specification is more flexible because it allows heterogeneous levels and trends across units, not just between treated and control groups.

10.5.4 The Parallel Trends Assumption

The identifying assumption for DiD is **parallel trends**: in the absence of treatment, treated and control units would have followed the **same trend** over time:

Y(0) denotes the untreated potential outcome. The intuition is: levels can differ (NJ may always have higher wages than PA), but **changes** should match. Both regions should move together absent the policy.

The assumption is **untestable**—it is a counterfactual statement about what would have happened without treatment. We cannot directly observe it. However, we can gain credibility by checking **pre-treatment** trends. If treated and control units move in parallel before treatment, we gain confidence (though no proof) that parallel trends hold post-treatment.

10.5.5 Checking Parallel Trends

Several methods help validate parallel trends:

Visual check: plot the two group means over time. Do they move in lockstep before treatment? Divergence suggests violation.

Placebo / event-study regression is more formal:

We estimate effects at all lags and leads relative to treatment. For pre-treatment periods, coefficients should be near zero (no pre-treatment effect). For post-treatment periods, we capture dynamic treatment effects.

Red flags include:
- Significant pre-trends with opposite sign from the estimated effect
- Divergence starting **before** the policy (suggesting selection or anticipation, not treatment effect)

If pre-trends diverge, **do not just add linear trend terms and hope**. Use a different method (synthetic control, matching on trends) instead.

10.5.6 Estimation Details

Several practical issues arise in DiD estimation:

Standard errors must be clustered at the **unit level**. Bertrand, Duflo, and Mullainathan (2004) showed that ignoring clustering overrejects the null by 40+ percent when autocorrelation is present. Most software has clustering options; always use them.

OLS with fixed effects gives the same point estimate as simple group means but allows covariates. Covariates help when they affect outcomes but are balanced across groups. They reduce variance without introducing bias.

Balanced vs. unbalanced panels: missing data or attrition biases estimates if non-random. Always report attrition rates and check whether balance is maintained across groups and time.

Common mistakes:
- Forgetting that treatment and time indicators enter as main effects, not just the interaction.
- Ignoring clustering when standard errors are autocorrelated.
- Using too short a pre-period, making it impossible to check parallel trends.

10.5.7 Robustness Checks

Strengthen DiD conclusions through robustness checks:

Placebo treatments in time: pretend treatment happened earlier. Should get no effect if the design is sound.

Placebo outcomes: apply DiD to unrelated outcomes that **should not** change. If minimum wage affects employment but not commodity prices, commodity prices should be unaffected in a DiD estimate.

Alternative control groups: use different comparison units. Results should be stable across reasonable choices. If the effect flips when you switch controls, the design is fragile.

Sensitivity to functional form: check whether results change with logarithmic vs. level outcomes, or linear vs. spline trend adjustments.

Falsification tests: subset the data in ways that **should not** yield effects. If they do, your design picks up something other than treatment.

The Card & Krueger minimum wage study exemplifies good robustness practice: they tried different fast-food chains and different controls, and the null effect held up, lending credibility.

10.5.8 Staggered Adoption and Modern Estimators

In reality, units are often treated at **different times** (staggered rollout). States adopt laws across different years; companies roll out products sequentially to different regions. Traditional **two-way fixed effects** (TWFE) models all these with:

However, recent econometric research (Goodman-Bacon 2021, de Chaisemartin & D'Haultfœuille 2020) revealed a critical flaw: with heterogeneous effects across units and time, TWFE becomes a **weighted average where some weights are negative**. Already-treated units serve as controls for later-treated units, which can produce wrong-sign bias.

Modern staggered DiD estimators fix this:

- **Callaway & Sant'Anna (2021)**: estimate group-time average treatment effects, ATT(g, t), for each cohort g at time t, then aggregate with clean weights.
- **Sun & Abraham (2021)**: interaction-weighted event-study estimator avoiding negative weights.
- **de Chaisemartin & D'Haultfœuille (2020)**: robust estimators under effect heterogeneity.
- **Borusyak, Jaravel, & Spiess (2024)**: imputation estimator using never-treated or not-yet-treated units as controls.

The key takeaway: for staggered adoption, **do not** default to TWFE. Use a modern estimator and report the full event-study plot showing effects by lag relative to treatment.

10.5.9 Summary: Strengths and Limits

Pros of DiD:
- Removes time-invariant confounders with unit fixed effects.
- Removes common shocks with time fixed effects.
- Uses both treated and control units, more credible than ITS alone.
- Flexible: extends to multiple periods, staggered adoption, covariates.

Cons of DiD:
- Parallel trends assumption is untestable—can only check pre-trends.
- Time-varying confounders still bias estimates.
- Staggered adoption with heterogeneous effects breaks naive TWFE.
- Standard errors require clustering—easy to get wrong.

Rule of thumb: DiD is credible when:
1. Pre-trends are visually parallel.
2. No co-occurring shocks hit treated and controls differently.
3. Staggered timing is handled with modern estimators.

10.6 Synthetic Control Methods

10.6.1 Motivation and Construction

Synthetic Control methods handle a scenario where DiD falls short: when **no single control unit** matches the treated unit. Studying California's 1988 tobacco control program (Prop. 99) illustrates the issue. California is large, diverse, and economically unique. Comparing it to any single state (Texas, Florida, or any other) is unconvincing because the state does not match California on key pre-treatment characteristics.

The **synthetic control idea** (Abadie & Gardeazabal 2003; Abadie, Diamond, & Hainmueller 2010) constructs a **weighted average** of control units that best matches the treated unit's pre-treatment characteristics. As shown in Figure 5, this weighted combination becomes the **synthetic control**—a data-driven, transparent counterfactual. We compare the treated unit to the synthetic control in the post-period. The gap is the estimated effect. The method's strength is transparency: reviewers can see **which** donors contribute and how much, making the counterfactual explicit.

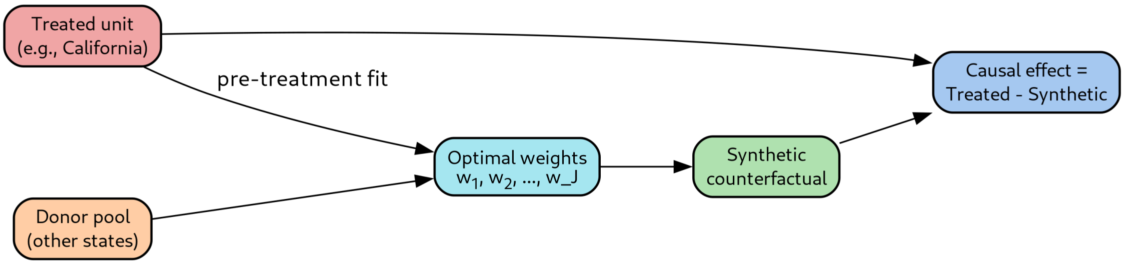

Figure 5: Synthetic control construction: the donor pool is weighted optimally

10.6.2 Formal Setup

Suppose we have J+1 units: unit 1 is treated at time t_start , units 2, …, J+1 are untreated (donors). Let X_1 be a vector of pre-treatment predictors for the treated unit, and X_0 is a matrix of the same predictors for donors (size k by J matrix).

Weights must satisfy: each weight is nonnegative and sums to one.

Non-negativity and unit-sum ensure weights are a **convex combination** of donors, placing the synthetic control in the **convex hull** of the donor pool. This prevents extrapolation—the synthetic control is never outside the range spanned by donors.

We choose optimal weights to minimize pre-treatment mismatch: The optimal weights minimize the squared pre-treatment mismatch between treated and donors. subject to weights being nonnegative and summing to one.

The predictor matrix V is diagonal, containing importance weights for each predictor, chosen to minimize mean-squared prediction error on pre-treatment outcomes. This is a nested optimization: the inner loop finds weights for a given predictor matrix; the outer loop finds V.

The **estimated causal effect** at time $t \geq t_0$ is the gap between the treated unit's outcome and the weighted average of donor outcomes.

10.6.3 Inference via Placebos

A critical challenge is that there is only **one** treated unit—standard inference does not apply. The solution is **placebo tests** by permutation:

1. Pretend each donor was the treated unit.
2. Compute the synthetic control and causal-effect trajectory for each.
3. Compare the **actual** treated unit's effect against this null distribution.

If the treated unit's effect is **larger in magnitude** than most placebo effects, we have evidence of a real effect. Compute an empirical p-value from the placebos.

A useful variant filters placebos to keep only those with good pre-treatment fit. A donor with poor pre-fit produces noise, not a legitimate null. In the California tobacco study, the post-period gap for California was larger than 38 of 39 placebo runs, yielding p approximately 0.026—strong evidence the policy worked.

10.6.4 When Synthetic Control Works

Synthetic control works well under specific conditions:

Long pre-treatment period: many time points allow matching on pre-treatment characteristics. Rule of thumb: at least 10+ observations.

Rich donor pool: many untreated units with diverse characteristics enable good pre-treatment fit via convex combinations.

Small number of treated units, usually one. Synthetic control is designed for aggregated units: countries, states, cities, or firms.

Outcome with moderate noise, low enough to detect signal above background variation.

Canonical successes include:
- Abadie & Gardeazabal (2003): effect of ETA terrorism on Basque GDP
- Abadie et al. (2010): California Prop. 99 tobacco control
- Abadie et al. (2015): German reunification's effect on West German GDP

Method	Control Group	Strength	Key Assumption
Granger	Own history	Predictive screen	Stationarity, no common causes
ITS	Pre-period of same unit	Works without controls	Stable pre-trend, no co-events
DiD	Untreated comparison	Removes time-invariant confounders	Parallel trends
Synthetic Control	Weighted donor pool	Data-driven counterfactual	Good pre-fit, no spillovers

Table 1: Comparison of causal inference methods for time series data.

10.6.5 When Synthetic Control Fails

Poor pre-treatment fit is a fatal flaw. If no convex combination of donors matches the treated unit, the treated unit is an **outlier** in donor space. Reported effects are unreliable—you are extrapolating in high-dimensional space.

Treatment contaminates donors (spillovers). If neighboring states are also affected by a policy, they are not valid controls. This violates the exclusion assumption.

Short pre-period or noisy outcome makes fit unreliable or prone to overfitting noise.

Confounding unobserved events near t_start masquerade as treatment effects. If California's economy experienced a shock around the tobacco law, synthetic control attributes it to the law.

Interpolation bias occurs when covariates are imbalanced: synthetic control averages over donors who differ in important but unmeasured ways.

Diagnostic: always plot the pre-treatment fit. Poor fit means do not trust the post-period gap.

10.6.6 Comparison with Other Methods

Each causal inference method has distinct strengths. Table 1 summarizes the key properties of each approach.

Granger Causality: uses the own history as control, offering fast predictive screening but only testing predictions, not interventions.

ITS: works without controls when policies are universal, but relies on stable pre-trends and extrapolation.

DiD: removes time-invariant confounders using unit fixed effects and common shocks using time fixed effects. Requires parallel pre-trends.

Synthetic Control: provides data-driven counterfactuals using weighted donors, transparent and avoids extrapolation (convex hull property), but inference is permutation-based and requires long, clean pre-periods.

Pros of synthetic control:
- Transparent weights—reviewers see which donors contribute.
- Avoids extrapolation (convex combination stays inside donor hull).
- Works when no single comparison unit is credible.

Cons of synthetic control:
- Inference requires placebos—less standard than standard statistical tests.
- Sensitive to donor pool choice—discipline needed.
- Requires long, clean pre-period.

10.6.7 Putting It All Together

Choose the appropriate method based on your data and design:

Use Granger causality when you want a fast predictive screen on two long time series and no intervention/experiment is available.

Use ITS when a single intervention has a sharp date and no comparison units exist, with a long and stable pre-period.

Use DiD when some units are treated and others are not, parallel pre-trends are plausible, and you have panel data with units and time.

Use Synthetic Control when one (or few) treated units have a rich donor pool, and no single donor is credible but a weighted combination is.

The key takeaway is that **temporal structure is a resource if exploited with the right tool, and a trap if ignored**. Each method has specific assumptions and conditions under which it excels. Matching method to design and checking assumptions carefully produces credible causal inference.